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Course: **Econometrics II**

Exercise Session: **1**
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Problem 1

Discuss the behavior of a Nadaraya-Watson mean regression estimate when one of i.i.d. observations, say (x_1, y_1), tends to assume extreme values. Specifically, discuss

(a) the case when x_1 is very big,

(b) the case when y_1 is very big.

Solution 1

(a) When x_1 is very big, we will have a big influence of this observation when estimating the regression in the right region of the support. With bounded kernels, we may find that no observations fall into the window. With unbounded kernels, the estimated regression line will go almost through (x_1, y_1).

(b) When y_1 is very big, the estimated regression line will have a hump in the region centered at x_1.

Problem 2

Firms produce some product using technology $f(l,k)$. The functional form of f is unknown, although we know that it exhibits constant returns to scale. For a firm i, we observe labor l_i, capital k_i, and output y_i, and the data-generating process takes the form $y_i = f(l_i, k_i) + \varepsilon_i$, where $\mathbb{E}[\varepsilon_i] = 0$ and ε_i is independent of (l_i, k_i). Using random sample $\{y_i, k_i, l_i\}_{i=1}^n$ suggest a nonparametric estimator of $f(l,k)$ which also exhibits constant returns to scale.

Solution 2

The CRS technology has the property that

$$f(l,k) = kf\left(\frac{l}{k}, 1\right).$$

Hence, the data-generating process is equivalent to

$$\frac{y_i}{k_i} = f\left(\frac{l_i}{k_i}, 1\right) + \frac{\varepsilon_i}{k_i},$$

from which follows that the Nadaraya-Watson estimator can be written as

$$\hat{f}\left(\frac{l}{k}, 1\right) = \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)\frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)} \Rightarrow \hat{f}(l,k) = k \cdot \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)\frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)}.$$

Problem 3

Recall the Nadaraya-Watson kernel estimator $\hat{g}(x)$ of the conditional mean $g(x) := \mathbb{E}[y|x]$ constructed for a random sample. Show that if $g(x) = c$, where c is some constant, then $\hat{g}(x)$ is unbiased.

Solution 3

Assume having a random sample $x_1, \dots, x_n$, and collect it as $\mathcal{X}$. Recall that

$$\hat{g}(x) = \frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}.$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\hat{g}(x)\right] &= \mathbb{E}\left[\mathbb{E}\left[\hat{g}(x)|\mathcal{X}\right]\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}|\mathcal{X}\right]\right] \\
&= \mathbb{E}\left[\frac{\sum_{i=1}^n \mathbb{E}\left[y_i|\mathcal{X}\right] K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] \\
&= c\mathbb{E}\left[\frac{\sum_{i=1}^n K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] = c.
\end{aligned}$$

First equality uses the law of iterated expectations, the second uses the definition of the $\hat{g}(x)$, the third follows because kernels are functions of $\mathcal{X}$, the forth follows from the definition of the conditional mean.

Problem 4

Suppose we have a random sample $\{x_i\}_{i=1}^n$ and let

$$\hat{F}(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}\{x_i \leq x\}$$

denote the empirical distribution function of x_i. Consider two density estimators:

- one-sided estimator:
$$\hat{f}_1(x) = \frac{\hat{F}(x+h) - \hat{F}(x)}{h},$$
- two-sided estimator:
$$\hat{f}_2(x) = \frac{\hat{F}(x+h/2) - \hat{F}(x-h/2)}{h}.$$

Show that:

(a) $\hat{F}(x)$ is an unbiased estimator of $F(x)$.

(b) The bias of $f_1(x)$ is $O(h^a)$. Find the value of a.

(c) The bias of $f_2(x)$ is $O(h^b)$. Find the value of b.

Solution 4

Please, consult the note on "big-o", "small-o" notation in case derivations are unclear.

(a) We show the unbiasedness using the fact that $\mathbb{E}\left[\mathbb{1}\{x_i \leq x\}\right] = F(x)$. Intuition is as follows: recall that $\mathbb{E}\left[\mathbb{1}\{x_i = x\}\right] = \mathbb{P}\{x_i = x\}$; intuitively, with inequality instead of equality those probabilities are summing up until the point x, which leads to equivalence with the cumulative distribution function.

Now, the unbiasedness,

$$\mathbb{E}\left[\hat{F}(x)\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n \mathbb{1}\{x_i \leq x\}\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}\left[\mathbb{1}\{x_i \leq x\}\right] = F(x).$$

(b) We will use the following Taylor approximation of $F(x+h)$ around x:

$$F(x+h) = F(x) + hf(x) + \frac{1}{2}h^2 f'(x) + o(h^2).$$

Substituting it, we have the bias

$$\begin{aligned}
\mathbb{E}\left[\hat{f}_1(x) - f(x)\right] &= \frac{1}{h}\left(F(x+h) - F(x)\right) - f(x) \\
&= \frac{1}{h}\left(F(x) + hf(x) + \frac{1}{2}h^2 f'(x) + o(h^2) - F(x)\right) - f(x) \\
&= \frac{1}{2}hf'(x) + o(h) \Rightarrow a = 1.
\end{aligned}$$

(c) Similarly to the point (b), we expand $F\left(x + \frac{h}{2}\right)$ and $F\left(x - \frac{h}{2}\right)$ around x:

$$F\left(x + \frac{h}{2}\right) = F(x) + \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) + \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3),$$

and

$$F\left(x - \frac{h}{2}\right) = F(x) - \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) - \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3).$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\hat{f}_2(x)\right] - f(x) &= \frac{1}{h}\left(F\left(x + \frac{h}{2}\right) - F\left(x - \frac{h}{2}\right)\right) - f(x) \\
&= \frac{1}{h}\left(F(x) + \ldots + o(h^3) - F(x) + \ldots + o(h^3)\right) \\
&= \frac{1}{h}\left(\frac{2}{6}\left(\frac{h}{2}\right)^3 f''(x)\right) + o(h^2) \\
&= \frac{1}{24}h^2 f''(x) + o(h^2) \Rightarrow b = 2.
\end{aligned}$$

Problem 5

Derive the asymptotic distribution of the Nadaraya-Watson estimator of the density of a scalar random variable x having a continuous distribution, similarly to how the asymptotic distribution of the Nadaraya-Watson estimator of the regression function is derived, under similar conditions. Give interpretation to how your expressions for asymptotic bias and asymptotic variance depend on the shape of the density.

Solution 5 (Bias)

Recall the Nadaraya-Watson density estimator, $\hat{f}(x) = \frac{1}{n}\sum_{i=1}^{n} K_h(x_i - x)$. During the first lecture you showed that $\mathbb{E}\left[\hat{f}(x)\right] = f(x) + o(1)$ as $h \to 0$. However, this result is not particularly revealing, since we do not see what causes the bias as h approaches (but not yet equal to) zero. To have a better picture, we work on the bias further taking a higher-order expansion of the density around x (the point we are estimating the density function at).

Now,

$$
\begin{aligned}
\mathbb{E}\left[\hat{f}(x) - f(x)\right] &= \int \frac{1}{h} K\left(\frac{x_i - x}{h}\right) f(x_i) dx_i - f(x) \\
&\overset{\left\{u = \frac{x_i - x}{h}\right\}}{=} \int K(u) f\left(x + hu\right) du - f(x) \\
&= \int K(u) \left(f(x) + f'(x)hu + \frac{1}{2} f''(x) h^2 u^2 + o(h^2) \right) du - f(x) \\
&= f(x) \int K(u) du + f'(x) h \int u K(u) du + \frac{1}{2} f''(x) h^2 \int u^2 K(u) du + o(h^2) - f(x) \\
&= \frac{1}{2} f''(x) h^2 \sigma_K^2 + o(h^2).
\end{aligned}
$$

In particular, we use here the fact that $\int K(u)du = 1$ due to normalization of the kernel, and $\int uK(u)du = 0$ due to the kernel symmetry.

What we see here is revealing: the bias is proportional to the second derivative of the density function. The second derivative, which describes the curvature of the density, is negative ($f''(x) < 0$) around the mode of the distribution, and is positive ($f''(x) > 0$) in the tails. That means that the density estimator will be downward-biased around the mode, and upward-biased around the tails. Also note that our results do not contradict the results you have obtained during the lecture: indeed, by putting $h \to 0$, we are getting rid of the bias.